

Bounce Rate Reconstruction from Available Vehicle Sensors

OU Kalman Filtering, Hybrid Learning, and Model-free Baselines

August 2026 Monthly Meeting

The target signal is unavailable at deployment

Deployment constraint

Bounce_rate_6D cannot be used as an input signal.

- ▶ Reconstruct the 100 Hz Bounce-rate waveform
- ▶ Use only signals already available on the vehicle
- ▶ Reserve the unavailable signal for training and evaluation labels
- ▶ Evaluate on completely held-out drivers

Group	Available signals	Count
IMU	Roll/yaw rate and 3-axis acceleration	5
Wheel speed	Four wheel-speed channels	4
Powertrain	Estimated and commanded torque	4
Total	Raw input channels	13

How much accuracy is possible without using Bounce rate at inference?

Driver-held-out evaluation prevents label leakage

2,429

Training episodes

Fitting and normalization

300

Validation episodes

Driver: Cho Hyun-seok
Early stopping

162

Held-out test episodes

Five unseen drivers

Common sequence setup

100 Hz sampling, 1,000 samples per episode, and 10 s observation windows. Short sequences are edge-padded; long sequences are truncated.

- ▶ Only recorded vehicle data were used; no synthetic data were generated.
- ▶ Test labels were excluded from fitting, normalization, early stopping, and checkpoint selection.
- ▶ Final metrics are per-episode correlation, RMSE, and temporal lag.

Four model families separate interpretability from context access

Family	Input at inference	Sequence model	Future samples
Model-based	Vertical acceleration	OU KF	No
Hybrid	KF states and innovation	OU KF + LSTM	No
Model-free online	13 sensor channels	LSTM, GRU, Transformer	No
Model-free offline	13 sensor channels	Bi-LSTM, Offline Transformer	Yes

Comparison boundary

KF and hybrid models use vertical acceleration only, whereas model-free models use all 13 channels. The final table is therefore a deployment comparison, not an equal-input architecture ablation.

OU KF adds a mean-reverting disturbance to a 1-DOF oscillator

State

$$\mathbf{x}(t) = [z(t) \quad v(t) \quad d(t)]^T$$

Continuous dynamics

$$\dot{z} = v, \quad \dot{v} = -\omega^2 z - 2\zeta\omega v + d + w_v$$

$$\dot{d} = -\lambda d + w_d$$

$$\mathbf{F} = \begin{bmatrix} 0 & 1 & 0 \\ -\omega^2 & -2\zeta\omega & 1 \\ 0 & 0 & -\lambda \end{bmatrix}$$

Ornstein-Uhlenbeck disturbance

$$\dot{d} = -\lambda d + w_d, \quad w_d \sim \mathcal{N}(0, q_f)$$

z latent vertical displacement

v latent vertical velocity

d OU disturbance acceleration

f natural frequency, $\omega = 2\pi f$

ζ damping ratio

q_v, q_f process-noise intensities

λ mean-reversion rate

r measurement-noise variance

Continuous-time dynamics are discretized before filtering

The OU model is defined in continuous time, while the data arrive at 100 Hz. With $\Delta t = 1/100 = 0.01$ s, define the continuous process-noise matrix $\mathbf{Q}_c = \text{diag}(10^{-10}, q_v, q_f)$ and form the Van Loan matrix

$$\mathbf{M} = \begin{bmatrix} \mathbf{F} & \mathbf{Q}_c \\ \mathbf{0} & -\mathbf{F}^\top \end{bmatrix} \Delta t.$$

$$e^{\mathbf{M}} = \begin{bmatrix} \mathbf{E}_{11} & \mathbf{E}_{12} \\ \mathbf{0} & \mathbf{E}_{22} \end{bmatrix}, \quad \mathbf{A} = \mathbf{E}_{11}, \quad \mathbf{Q} = \mathbf{E}_{12} \mathbf{A}^\top.$$

Discrete model used by the KF

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad y_k = \mathbf{C}\mathbf{x}_k + v_k.$$

This preserves the continuous-time dynamics and process covariance over one sampling interval; it is not an Euler approximation. The implementation stores the resulting discrete $\mathbf{Q}$ directly.

One acceleration channel updates all three states online

Linear-Gaussian model using the lecture notation

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{G}\mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad y_t = \mathbf{C}\mathbf{x}_t + v_t, \quad v_t \sim \mathcal{N}(0, \mathbf{R})$$

$$y_t = 9.81(\text{IMU_VerAccelVal}_t - 1), \quad \mathbf{C} = \begin{bmatrix} -\omega^2 & -2\zeta\omega & 1 \end{bmatrix}, \quad \mathbf{R} = [r]$$

Prediction

$$\hat{\mathbf{x}}_{t+1|t} = \mathbf{A}\hat{\mathbf{x}}_{t|t}$$

$$\mathbf{P}_{t+1|t} = \mathbf{A}\mathbf{P}_{t|t}\mathbf{A}^\top + \mathbf{G}\mathbf{Q}\mathbf{G}^\top$$

Innovation

$$\nu_{t+1} = y_{t+1} - \mathbf{C}\hat{\mathbf{x}}_{t+1|t}$$

$$\mathbf{S}_{t+1} = \mathbf{C}\mathbf{P}_{t+1|t}\mathbf{C}^\top + \mathbf{R}$$

Correction

$$\mathbf{K}_{t+1} = \mathbf{P}_{t+1|t}\mathbf{C}^\top\mathbf{S}_{t+1}^{-1}$$

$$\hat{\mathbf{x}}_{t+1|t+1} = \hat{\mathbf{x}}_{t+1|t} + \mathbf{K}_{t+1}\nu_{t+1}$$

$$\begin{aligned} \mathbf{P}_{t+1|t+1} &= (\mathbf{I} - \mathbf{K}_{t+1}\mathbf{C})\mathbf{P}_{t+1|t}(\mathbf{I} - \mathbf{K}_{t+1}\mathbf{C})^\top \\ &\quad + \mathbf{K}_{t+1}\mathbf{R}\mathbf{K}_{t+1}^\top \end{aligned}$$

The implementation obtains $\mathbf{A}$ and $\mathbf{G}\mathbf{Q}\mathbf{G}^\top$ at $\Delta t = 0.01$ s by Van Loan discretization and stores the latter as its discrete $\mathbf{Q}$ matrix. Bounce rate is calibrated as $\hat{b}_t = g\hat{v}_{t|t} + c$ using training labels only.

The KF parameters optimize reconstruction, not physical identification

Fitting protocol

- ▶ Up to 300 uniformly sampled training episodes
- ▶ Powell optimization, at most 40 iterations
- ▶ Objective: normalized RMSE after affine calibration

Parameter	Search range	Fitted
f	0.5–8.0 Hz	4.0043
ζ	0.05–5.0	0.2990
q_v	e^{-12} – e^3	1.11×10^{-3}
q_f	e^{-12} – e^5	148.4132
r	e^{-12} – e^4	0.3306
λ	0.05–100	13.4850

Held-out result

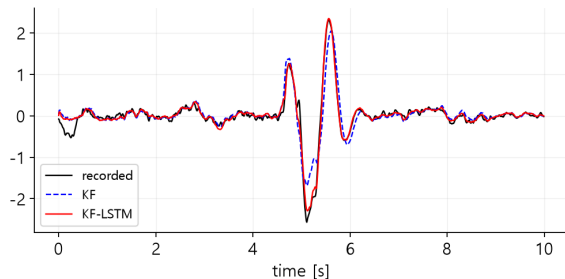
Median correlation	0.9234
Median RMSE	0.2557
Median lag	10 ms

Interpretation limit

The expanded range moved f above the former 3 Hz cap. These fitted values are effective reconstruction parameters, not identified suspension properties.

The model is useful as an online reconstruction baseline and as a structured feature generator.

A causal LSTM learns only the KF residual



The correction improves both peaks and low-amplitude regions.

$$\hat{b}_k^{\text{hybrid}} = \hat{b}_k^{\text{KF}} + \Delta \hat{b}_k^{\text{LSTM}}$$

$$\mathbf{u}_k = [\hat{z}_k, \hat{v}_k, \hat{d}_k, \nu_k]^T$$

LSTM	2 layers, hidden 64
Head	64 $\rightarrow$ 32 $\rightarrow$ 1, SiLU
Parameters	52,833
Optimizer	Adam, LR 10^{-3}
Batch / epochs	32 / 30 max
Early stopping	Patience 8

Median test result

Corr. 0.9828; RMSE 0.1226; lag 0 ms.

RMSE decreases by 52.0% versus KF alone.

All model-free networks use the same supervised protocol

Direct sequence mapping

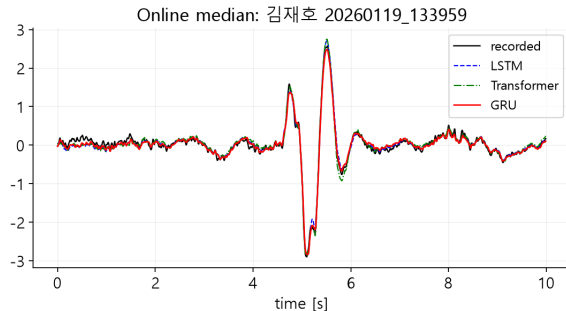
$$\hat{b}_{1:T} = f_{\phi}(\mathbf{s}_{1:T}), \quad \mathbf{s}_t \in \mathbb{R}^{13}$$

- ▶ Per-channel input z-score from training data
- ▶ Scalar target mean and standard deviation from training data
- ▶ Sample-wise MSE on the normalized target
- ▶ Lowest validation-MSE checkpoint retained

Setting	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-4}
Maximum epochs	30
LR scheduler	Plateau, factor 0.2, patience 3
Early stopping	Patience 8
Gradient clipping	Norm 5.0
Mixed precision	CUDA only

Online models passed a future-input perturbation test; offline models intentionally use the full episode.

GRU is the strongest model-free online candidate



All three models use current and past samples only.

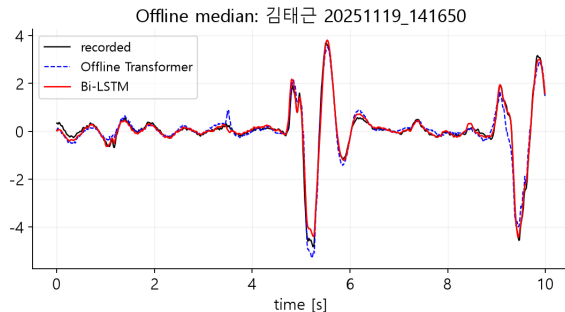
Model	Backbone	Params.
LSTM	2-layer uni., $h = 64$	53,569
GRU	2-layer uni., $h = 64$	40,193
Transformer	3L, 4H, $d = 64$	165,505

Model	Batch	Epoch	Corr.	RMSE
LSTM	32	28	0.9908	0.0900
GRU	32	29	0.9918	0.0841
Transformer	8	30	0.9863	0.1081

Online choice

GRU achieves the lowest error with the fewest parameters.

Bi-LSTM is best overall, but requires the complete episode



Future context is a capability, not a guarantee of higher accuracy.

Model	Backbone	Params.
Bi-LSTM	2-layer bi., $h = 64$	139,905
Offline T.	3L, 4H, $d = 64$	165,505

Model	Batch	Epoch	Corr.	RMSE
Bi-LSTM	32	27	0.9927	0.0790
Offline T.	8	29	0.9819	0.1203

Offline choice

Bi-LSTM improves median RMSE by 0.0051 over online GRU.

The deployment boundary determines the practical shortlist

Role	Model	Main trade-off	Corr.	RMSE
Model-based	KF	One sensor; explicit states	0.9234	0.2557
Hybrid	KF + LSTM	One sensor; learned correction	0.9828	0.1226
Online	GRU	Best real-time accuracy	0.9918	0.0841
Offline	Bi-LSTM	Best overall; future context	0.9927	0.0790

Current evidence

Fully model-free recurrent models provide the best accuracy on this held-out split.

Before final selection

Repeat driver splits and seeds, measure inference latency, and run sensor ablation under an equal-input protocol.